

A Memory construction

Committed events
daily buffer of the open local day



day closes

LLM Trajectory extraction
segments with typed facts and citations

18:22–18:48 walking steps 2,310 [e_{17}, e_{18}]
✓ facts $\subseteq$ cited events $\Rightarrow$ kept in W_t
✗ “walked 5 km” has no source fact $\Rightarrow$ rejected

7 daily records

LLM Regularity proposal
weekly, over 7 completed records

trigger weekday, low activity after 17:30
expect walk within 18:00–19:00
replay **M T W T F S S** weekend excl.
 $n_f=4, n_v=1, \text{conf } 0.80, \text{lift} \geq 1 \Rightarrow$ **active**

ProactMem

Event memory
 S_t
last 72 h,
 ≤ 99 events

Temporal trajectories
 W_t
source-checked
segments

Behavioral regularities
 L_t
active: visible
candidate: hidden

B Event-driven decision

online, per event

Context C_t

S_t recent events

Q_t episode state

W_t trajectories

$C_t = (e_t, S_t, W_t, R_t, Q_t)$

Retrieval top 3

✓ active regularity
✓ trigger + time match
✓ not yet fulfilled

e_t 18:30, low activity

R_t regularities

e_t current event

Decision LLM

$(\tilde{y}_t, \tilde{m}_t) = \pi_\theta(C_t)$
timely, feasible,
low-risk service?

Intervention control

compare with allowed
interventions;
✗ duplicate $\rightarrow$ abstain

INTERVENE

“Up for a short
walk?”

ABSTAIN
otherwise

commit e_t after the decision is finalized; it affects only later decisions.